

# Point Neuron Embedded Kalman Filter for Narrowband Sound Source Tracking

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**Abstract**—This paper proposes the Point Neuron Embedded Kalman Filter (PNEKF), a novel Kalman Filter variant for tracking multiple narrowband sound sources in reverberant environments. Unlike existing methods that rely on discrete, grid-based source localization in each time frame, PNEKF requires localization only at initialization and thereafter operates directly on raw microphone signals by integrating the Point Neuron framework with Kalman filtering. This enables efficient, continuous 3-D tracking. We benchmark the performance of PNEKF against conventional tracking algorithms across a range of frequencies and source trajectories for the single-source case. The results demonstrate that PNEKF consistently achieves higher tracking accuracy than the baselines under all tested conditions. Finally, we demonstrate the robustness of the framework by extending it to track multiple sources in reverberant environments.

**Index Terms**—Sound source tracking, reverberant environments, Kalman Filter, Point Neuron Learning

## I. INTRODUCTION

Sound source tracking (SST) is crucial for spatial awareness of acoustic events in a multitude of applications. These include Audio Augmented Reality (AAR), smart conferencing [1], where it improves perceived realism by preserving spatial consistency; surveillance [2], where it enables the detection and monitoring of suspicious or unauthorized acoustic events; and robotics[3], where it enables robots to follow a person of interest, enhancing functions such as speech recognition. Although broadband sources are common in many practical scenarios, narrowband SST remains highly relevant in applications involving machine noises, musical instruments, wireless communications and resource-constrained systems, where accurate frequency-specific localization is required. Prior work on SST has largely focused on combining localization techniques [4] with filtering or smoothing methods. Multiple Signal Classification (MUSIC)-based localization has been paired with Kalman filtering through generalised eigenvalue decomposition [5]. Time Difference of Arrival (TDOA) estimates from generalised cross-correlation have been employed with adaptive distributed particle filters [6], conventional particle filters [7], and multi-model Kalman filters [8]. Steered-response-power beamforming has likewise been integrated with particle filters [9] or with diagonally unloaded beamforming followed by Kalman filtering [10]. For multi-source tracking, notable approaches include the TRAMP algorithm [11] and sequential Monte Carlo probability-hypothesis density (SMC-PHD) filtering with MUSIC-based localization [12].

Despite these advances, existing approaches suffer from two key limitations. First, most existing tracking algorithms rely

on grid-based sound source localization (SSL) estimates at every time frame, leading to grid mismatch errors. A recent continuous-domain gradient method [13] uses implicit smoothing rather than an explicit motion model, suiting only smooth, moderate-speed motion. Second, narrowband source tracking in reverberant environments is hindered by limited frequency content and reflected image sources. Three-dimensional (3-D) tracking of multiple moving sources under such conditions has received limited attention.

The primary contributions of our paper are:

- We introduce the Point Neuron Embedded Kalman Filter (PNEKF), a novel variant of Kalman filtering (KF) that integrates the point neuron framework [14] to enable 3-D tracking of multiple narrowband sound sources.
- We performed simulations across a range of frequencies and trajectories to evaluate the proposed method, benchmarking its performance against baselines in the single-SST case.
- We extend the PNEKF framework to multiple-SST, demonstrating its capability to simultaneously track multiple moving sources with significantly improved accuracy compared to baseline methods.

Unlike previous methods, PNEKF uses SSL only once for initialization and thereafter operates directly on microphone measurements, enabling continuous 3-D tracking while avoiding grid mismatch errors. The choice of point neurons as virtual sources was motivated by two key advantages: (1) complete differentiability, allowing efficient backpropagation for optimization, and (2) clear physical interpretation of weights and biases as acoustic parameters, enabling natural integration with our SST problem formulation. By embedding point neurons along room boundaries, PNEKF dynamically simulates image sources induced by reverberation, enabling robust tracking in highly reverberant environments. The empirical evaluation shows that PNEKF outperforms all tested baselines for single-SST across a range of frequencies and can also be robustly scaled to track multiple moving narrowband sound sources in 3-D accurately.

## II. PROBLEM FORMULATION

**Setup:** Consider  $N$  narrowband sound sources of frequency  $f$  with time-varying signal strengths that move freely within a reverberant room. A set of  $Q$  microphones is placed along the walls, floor, and ceiling as an empirical design choice to measure sound pressure across  $T$  time frames,  $t \in \{1, 2, \dots, T\}$ .

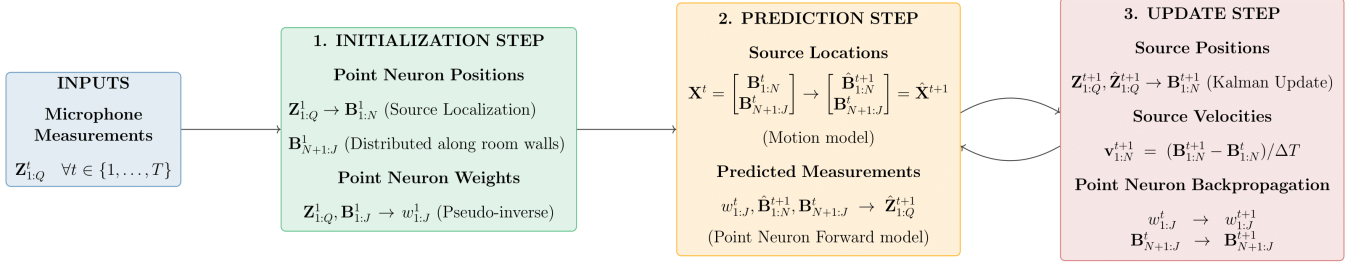


Fig. 1: Flowchart of the proposed PNEKF algorithm for SST. PNEKF initializes point neuron weights and positions at  $t = 1$  using SSL. It then alternates between prediction and update steps, operating directly on microphone measurements. In prediction, source locations are propagated via a linear motion model and predicted measurements are computed using (9). In update, Kalman update estimates source positions, while velocities are computed from successive source positions, and backpropagation updates weights and positions using (15).

The position of the  $n^{\text{th}}$  source at time  $t$  is denoted by  $\mathbf{s}_n^t = (x_n^t, y_n^t, z_n^t)$  for  $n \in \{1, 2, \dots, N\}$ , while the position of the  $q^{\text{th}}$  microphone is  $\mathbf{m}_q = (x_q, y_q, z_q)$  for  $q \in \{1, 2, \dots, Q\}$ .

**Point Neurons:** To capture the effects of reverberation and facilitate SST, we adopt the point neuron framework [14], in which  $J$  virtual sources termed as point neurons reconstruct the sound pressure field at the microphones. The first  $N$  point neurons are placed at the estimated positions of the moving sources to capture the direct component, while the remaining  $J - N$  neurons are uniformly distributed along the walls, floor, and ceiling to model the reverberant component. This arrangement follows the classical sound field decomposition [15], in which any reverberant sound field can be expressed as the sum of its particular and homogeneous solutions, corresponding to the direct and reverberant components, respectively. At time frame  $t$ , the  $J$  point neurons are parameterized by weights  $w_j^t$  and biases  $\mathbf{B}_j^t \in \mathbb{R}^3$  for  $j \in \{1, \dots, J\}$ , corresponding to the strength and position of the virtual source, respectively. Their resulting sound pressure  $P_q^t(f)$  at the  $q^{\text{th}}$  microphone is:

$$P_q^t(f) = \sum_{j=1}^J w_j^t \cdot \text{PN}(\mathbf{B}_j^t, \mathbf{m}_q, f) \quad (1)$$

$$= \sum_{j=1}^J w_j^t \cdot \|\mathbf{B}_j^t\|_2 e^{-ik\|\mathbf{B}_j^t\|_2} \frac{e^{ik\|\mathbf{m}_q - \mathbf{B}_j^t\|_2}}{4\pi\|\mathbf{m}_q - \mathbf{B}_j^t\|_2},$$

where  $k = 2\pi f/c$  is the wave number,  $c$  is the speed of sound,  $\|\cdot\|_p$  denotes the  $\ell_p$  norm, and  $\text{PN}(\mathbf{B}_j^t, \mathbf{m}_q, f)$  represents the nonlinear forward response of the  $j$ -th point neuron at microphone  $q$  at time  $t$ . The forward model matrix of Point Neuron Learning,  $\mathbf{D}^t \in \mathbb{C}^{Q \times J}$ , stacks these responses for all microphone–neuron pairs at each timestep.

**Objective:** We address the problem of tracking the 3-D positions of  $N$  moving narrowband sound sources in a reverberant environment. The estimated location of the  $n^{\text{th}}$  source at time  $t$  given by  $\hat{\mathbf{s}}_n^t$  is modelled by the position of its corresponding point neuron  $\mathbf{B}_n^t$ :

$$\hat{\mathbf{s}}_n^t = \mathbf{B}_n^t, \quad n \in \{1, 2, \dots, N\}. \quad (2)$$

**System Model:** At time frame  $t$ , the state space  $\mathbf{X}^t$  consists of the estimated positions  $\mathbf{B}_{1:N}^t$  and velocities  $\mathbf{v}_{1:N}^t \in \mathbb{R}^3$  of

the  $N$  moving sound sources, while the measurement space  $\mathbf{Z}^t$  corresponds to the sound pressure field recorded by  $Q$  microphones:

$$\mathbf{X}^t = [\mathbf{B}_1^t, \mathbf{B}_2^t, \dots, \mathbf{B}_J^t, \mathbf{v}_1^t, \mathbf{v}_2^t, \dots, \mathbf{v}_N^t]^\top, \quad (3)$$

$$\mathbf{Z}^t = [P_1^t(f), P_2^t(f), \dots, P_Q^t(f)]^\top. \quad (4)$$

Process noise from source accelerations and measurement noise due to microphone uncertainty are modelled as Gaussian random variables  $y^t$  and  $r^t$  with covariance matrices  $\mathcal{Y}$  and  $\mathcal{R}$ , respectively.

### III. POINT NEURON EMBEDDED KALMAN FILTER

The PNEKF algorithm has 4 steps as shown in Fig. 1.

#### A. Initialization Step

The tracking process begins with the initialization of the weights and positions of the point neurons at  $t = 1$ .

**Positions:** To initiate the tracking process, the sound sources are first localized using an SSL algorithm, and the  $N$  point neurons corresponding to the sources are placed at these estimated positions  $\mathbf{B}_{1:N}^1$ . The remaining  $J - N$  point neurons  $\mathbf{B}_{N+1:J}^1$  are uniformly distributed along planes aligned with the room boundaries.

**Weights:** Given the microphone measurements and point neuron positions, the weights  $w_{1:J}^1$  can be directly computed using the Moore-Penrose pseudoinverse of the  $\mathbf{D}^1$  matrix.

$$[w_1^1, \dots, w_J^1]^\top = (\mathbf{D}^1)^\dagger \mathbf{Z}^1, \quad (5)$$

where  $(\cdot)^\dagger$  denotes the Moore-Penrose pseudoinverse.

#### B. Prediction Step

The prediction step of PNEKF employs a linear motion model. The predicted next state  $\hat{\mathbf{X}}^{t+1}$  and state error covariance  $\hat{\mathbf{F}}^{t+1}$  are obtained from the standard Kalman filter prediction equations [16], [17].

$$\hat{\mathbf{X}}^{t+1} = \mathbf{A}\mathbf{X}^t + \mathbf{C}y^t, \quad (6)$$

$$\hat{\mathbf{F}}^{t+1} = \mathbf{A}\mathbf{F}^t\mathbf{A}^\top + \mathcal{Y}. \quad (7)$$

We define the state transition and noise matrices,  $\mathbf{A}$  and  $\mathbf{C}$ , as in [8].

$$\mathbf{A} = \begin{bmatrix} \mathbf{I}_N & \mathbf{0} & \Delta T \mathbf{I}_N \\ \mathbf{0} & \mathbf{I}_{J-N} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I}_N \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} \frac{1}{2}(\Delta T)^2 \mathbf{I}_N \\ \mathbf{0} \\ \Delta T \mathbf{I}_N \end{bmatrix}.$$

Here,  $\mathbf{I}_N$  denotes the  $N \times N$  identity matrix,  $\mathbf{0}$  a zero matrix of appropriate dimensions, and  $\Delta T$  the discrete time step between successive frames. The matrices  $\mathbf{A}$  and  $\mathbf{C}$  are designed to encode the dynamics of the point neurons: the reverberant point neuron locations  $\mathbf{B}_{N+1:J}^t$  are modelled as static, while source point neurons  $\mathbf{B}_{1:N}^t$  evolve with velocity  $\mathbf{v}_{1:N}^t$ , with both their positions and velocities perturbed by acceleration. Accordingly,  $\mathbf{A}$  determines the evolution of source positions and velocities while maintaining the reverberant states fixed, and  $\mathbf{C}$  introduces acceleration noise to account for deviations from the constant velocity assumption.

**Predicted Measurements:** Conventional KF [18] computes predicted measurements ( $\hat{\mathbf{Z}}^{t+1}$ ) by linearizing the relation between measurement and state space using the Jacobian  $\mathbf{H}^t$ .

$$\hat{\mathbf{Z}}^{t+1} = \mathbf{H}^t \hat{\mathbf{X}}^{t+1}, \quad (8)$$

where  $\mathbf{H}^t$  is given by:

$$\mathbf{H}^t = \begin{bmatrix} \frac{\partial P_1^t(f)}{\partial \mathbf{B}_1^t} & \dots & \frac{\partial P_1^t(f)}{\partial \mathbf{B}_J^t} & \frac{\partial P_1^t(f)}{\partial \mathbf{v}_1^t} & \dots & \frac{\partial P_1^t(f)}{\partial \mathbf{v}_N^t} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial P_Q^t(f)}{\partial \mathbf{B}_1^t} & \dots & \frac{\partial P_Q^t(f)}{\partial \mathbf{B}_J^t} & \frac{\partial P_Q^t(f)}{\partial \mathbf{v}_1^t} & \dots & \frac{\partial P_Q^t(f)}{\partial \mathbf{v}_N^t} \end{bmatrix}.$$

However, PNEKF employs the full non-linear forward model of Point Neuron Learning [14] in (1), to map the predicted point neuron locations  $\hat{\mathbf{B}}_{1:J}^{t+1}$ , microphone positions  $\mathbf{m}_q$ , and point neuron weights  $w_j^t$  to the estimated sound pressure.

$$\hat{\mathbf{Z}}_q^{t+1} = \sum_{j=1}^J w_j^t \cdot \text{PN}(\hat{\mathbf{B}}_j^{t+1}, \mathbf{m}_q, f). \quad (9)$$

Since the measurements are complex-valued, we use their real-valued augmented forms, denoted by  $\mathbf{Z}^t$ ,  $\hat{\mathbf{Z}}^t$ , and  $\mathbf{H}^t$ .

$$\mathbf{M} = [\Re(\mathbf{M}) \ \Im(\mathbf{M})], \quad \mathbf{M} \in \{\mathbf{Z}^t, \hat{\mathbf{Z}}^t, \mathbf{H}^t\}. \quad (10)$$

#### C. Update Step

We compute the Kalman gain  $\mathbf{K}^t \in \mathbb{R}^{3(J+N) \times 2Q}$  [16], [17] to balance predicted and observed measurements.

$$\mathbf{K}^t = \hat{\mathbf{F}}^t (\mathbf{H}^t)^\top \left( \mathbf{H}^t \hat{\mathbf{F}}^t (\mathbf{H}^t)^\top + \mathbf{R} \right)^{-1}. \quad (11)$$

**Position:** The estimated positions of the  $N$  moving sound sources are obtained using the Kalman update rule [16], [17] to the  $N$  point neurons corresponding to the sources  $\mathbf{B}_{1:N}^t$ .

$$\mathbf{B}_{1:N}^{t+1} = \hat{\mathbf{B}}_{1:N}^{t+1} + \mathbf{K}_{[1:3N,:]}^t (\mathbf{Z}^{t+1} - \hat{\mathbf{Z}}^{t+1}). \quad (12)$$

**Velocities:** Since measured sound pressure depends only on source positions, velocities are not directly observable and are instead computed implicitly from successive position estimates using the Kalman update rule.

$$\mathbf{v}_{1:N}^{t+1} = \frac{(\mathbf{B}_{1:N}^{t+1} - \mathbf{B}_{1:N}^t)}{\Delta T}. \quad (13)$$

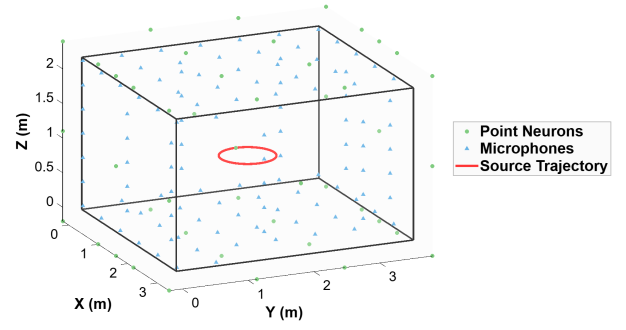


Fig. 2: Simulation setup of a  $3.6 \times 3.2 \times 2.2$  m room with reverberation coefficients  $[0.7, 0.8, 0.65, 0.7, 0.5, 0.6]$  and maximum reflection order 10. Microphone placements, point neuron locations, and a representative source trajectory are illustrated. The number and trajectories of the sources vary across experiments.

**Backpropagation:** The positions of the  $J - N$  reverberant point neurons and the weights of all the  $J$  point neurons are updated using the backpropagation step of Point Neuron Learning [14], while the position of the point neurons corresponding to sources remains fixed. To evaluate the point neuron outputs for these updates, the expected microphone measurement  $\mathcal{Z}_q^{t+1}(f)$  is computed using the updated source locations  $\mathbf{B}_{1:N}^{t+1}$ , the current reverberant point neuron locations  $\mathbf{B}_{N+1:J}^t$ , and the current point neuron weights  $w_{1:J}^t$ :

$$\mathcal{Z}_q^{t+1}(f) = \sum_{j=N+1}^J w_j^t \text{PN}(\mathbf{B}_j^t, \mathbf{m}_q, f) + \sum_{j=1}^N w_j^t \text{PN}(\mathbf{B}_j^{t+1}, \mathbf{m}_q, f). \quad (14)$$

The backpropagation loss function is then defined as:

$$\mathcal{L}(w_{1:J}^t, \mathbf{B}_{N+1:J}^t) = \lambda \sum_{j=1}^J \|w_j^t\|_1 + \sum_{q=1}^Q \left\| \left( \mathcal{Z}_q^{t+1}(f) - P_q^{t+1}(f) \right) \right\|_2^2. \quad (15)$$

The first term enforces weight sparsity, and the second minimizes the sound field reconstruction error at the microphones.

## IV. EXPERIMENTAL RESULTS

This section presents a simulation-based evaluation of the proposed PNEKF algorithm using the tracking error metric defined in Section IV-B. Section IV-C compares its performance with baseline methods in the single-source case and demonstrates its robustness by extending the framework to multiple-source tracking.

#### A. Simulation Setup

We simulate a rectangular room of dimensions  $3.6 \text{ m} \times 3.2 \text{ m} \times 2.2 \text{ m}$ , by using the image-source method [19] to compute the acoustic transfer function between each source and each microphone at every time frame  $t$ . The reverberation coefficients are set to  $[0.7, 0.8, 0.65, 0.7, 0.5, 0.6]$ , and the maximum reflection order is fixed at 10. To record the sound field, we deploy  $Q = 95$  microphones uniformly along the

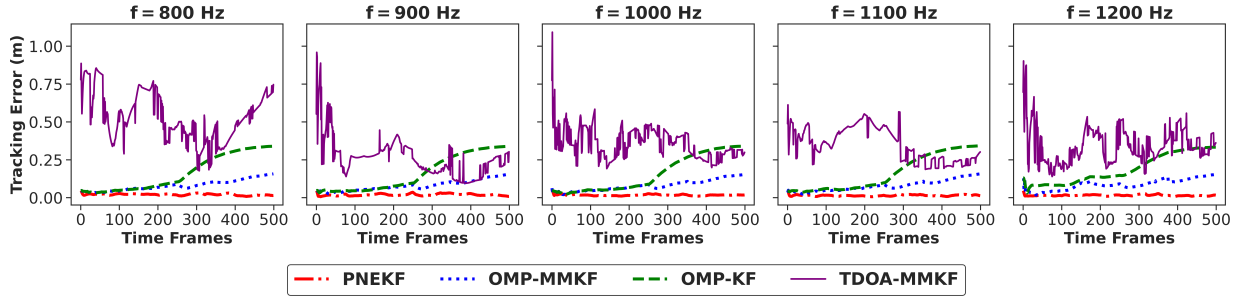


Fig. 3: Comparison of the proposed PNEKF with TDOA-MMKF, OMP-MMKF, and OMP-KF in the single-source tracking scenario, evaluated using the metric defined in Section IV-B with  $Q = 95$  microphones. The PNEKF consistently outperforms all baselines across the tested source frequencies  $f \in \{800, 900, 1000, 1100, 1200\}$  Hz, maintaining near-zero tracking error.

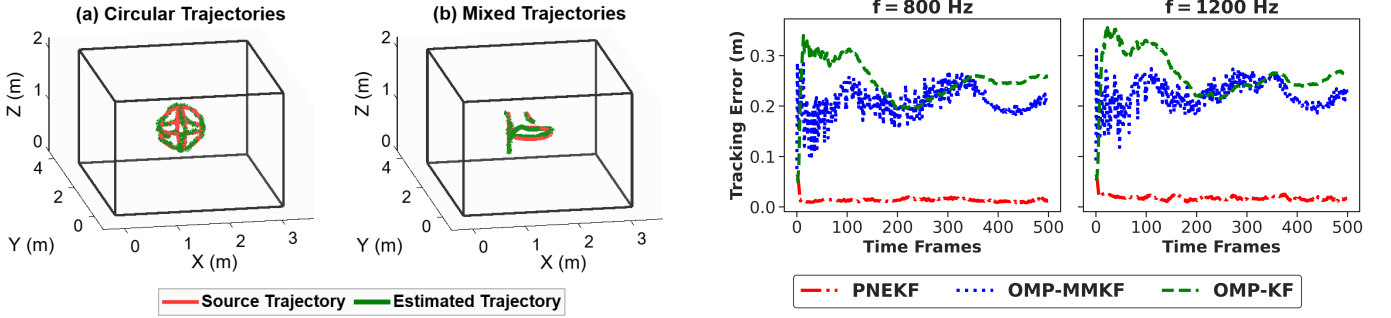


Fig. 4: Multiple-source tracking results using PNEKF with  $Q = 119$  microphones. Left: Qualitative visualization of PNEKF tracking performance at  $f = 1200$  Hz for three sources following circular and mixed trajectories. Right: Quantitative comparison with OMP-MMKF and OMP-KF at  $f = 800$  Hz and  $f = 1200$  Hz using average tracking error across all sources, evaluated per the metric in Section IV-B across both trajectories. The proposed PNEKF can thus accurately track multiple source trajectories within the ROI, demonstrating its scalability.

walls, floor, and ceiling, as in Fig. 2, each operating at a signal-to-noise ratio (SNR) of 20 dB. The moving sound sources are constrained to a  $1\text{ m} \times 1\text{ m} \times 1\text{ m}$  Region of Interest (ROI) at the centre of the room. All simulations are conducted in the frequency domain, with each source trajectory discretized into  $T = 500$  spatial points corresponding to each time frame, separated by  $\Delta T = 1\text{ s}$ . Experiments are conducted with  $N$  ranging from 1 to 4 sources, where each source operates at a frequency selected from  $[800, 900, 1000, 1100, 1200]$  Hz and has a weight sampled uniformly from  $[0.5, 2.5]$  at each time instant. To account for reverberation,  $J - N = 78$  point neurons are placed on planes parallel to the room boundaries (walls, floor, and ceiling), with a 0.2 m outward offset chosen for numerical stability, since point neurons cannot coincide with microphones (1). In our experiments, we empirically determine the initialization of the diagonal error covariance matrices as:  $\mathbf{F}^1$  with variances of  $10^{-6}$  for positions and  $10^{-5}$  for velocities,  $\mathbf{R}$  with variance of  $10^{-2}$  for measurements, and  $\mathbf{Y}$  with variance of  $10^{-10}$  for process noise.

**Source localization:** To initiate the tracking process, we first locate the sound sources in the room by applying a modified Orthogonal Matching Pursuit (OMP) algorithm [20], [21] that leverages point neurons distributed along the walls for dereverberation and a Cartesian grid of candidate point neurons inside the ROI to iteratively estimate source positions

from the dereverberated measurements.

#### B. Evaluation Metric

To measure the estimation accuracy of our SST algorithm, we define an evaluation metric based on the Root Mean Square Error (RMSE) between the estimated  $\hat{\mathbf{s}}_n^{t,(m)}$  and ground-truth source locations  $\mathbf{s}_n^{t,(m)}$ . At each time instant  $t$ , the tracking error  $\mathbf{TE}_n(t)$  for source  $n \in \{1, \dots, N\}$ , averaged over all  $M$  experiments, is given by:

$$\mathbf{TE}_n(t) = \sqrt{\frac{1}{M} \sum_{m=1}^M \|\hat{\mathbf{s}}_n^{t,(m)} - \mathbf{s}_n^{t,(m)}\|_2^2}. \quad (16)$$

#### C. Results and Discussion

We evaluate the proposed PNEKF algorithm in the single-source tracking scenario ( $N = 1$ ) using  $M = 3$  representative trajectories—linear, circular, and zigzag—and compare its performance against three baselines: TDOA-based localization [22] with Multi-Model KF (TDOA-MMKF) [8], modified OMP-based localization with Multi-Model KF (OMP-MMKF), and modified OMP-based localization with conventional KF (OMP-KF). Traditional SSL methods often fail under strong reverberation and limited frequency content, motivating the use of our modified OMP algorithm as the localization method in the latter two baselines. In both TDOA-MMKF and OMP-MMKF, the MMKF employs three motion

models for the prediction step: stationary, constant velocity, and constant acceleration, which allows greater adaptability to source dynamics compared to conventional KF. Fig. 3 shows that the proposed PNEKF algorithm consistently outperforms all baselines across these trajectories, maintaining near-zero tracking error throughout. This performance can be attributed to PNEKF requiring SSL estimates only once during initialization and thereafter operating directly on microphone measurements in the continuous spatial domain, thereby avoiding the grid mismatch errors that arise in baselines such as TDOA-MMKF, OMP-MMKF, and OMP-KF, which depend on grid-based SSL at every time frame.

**Multiple Sound Source Tracking:** To demonstrate the scalability of our framework, we extend it to the case of multiple SSTs. Evaluations with  $N = 3$  sources under the same setup show that the PNEKF algorithm can accurately track multiple sources within the ROI, requiring only a modest increase in the number of microphones to  $Q = 119$ . Although experiments were conducted for  $f \in \{800, 900, 1000, 1100, 1200\}$  Hz, Fig. 4 shows selected results at  $f = 800$  and  $1200$  Hz for brevity. The qualitative visualization demonstrates accurate trajectory estimation, while the quantitative comparison with OMP-MMKF and OMP-KF baselines shows that PNEKF consistently achieves lower tracking errors. These results demonstrate the scalability of PNEKF for multiple SST.

**Limitations:** Although our setup employs a relatively large number of microphones, the count is comparable to that used in existing narrowband localization frameworks [21], while remaining significantly smaller than the sound-field dimensionality of the ROI. For the simulated  $1 \text{ m}^3$  ROI (with maximum radius  $r_{\max} \approx 0.71 \text{ m}$ ), the sound-field dimensionality  $[(k_{\max} r_{\max} + 1)^2]$  [23] evaluates to approximately 281, where  $k_{\max}$  is the maximum wavenumber across all tested frequencies. This substantially exceeds the number of sensors used, despite the reverberant conditions. Furthermore, embedding microphones in room boundaries reduces practical deployment constraints, and the exact bounds on the number of microphones needed for moving sources are an open problem.

## V. CONCLUSION

We proposed PNEKF, a novel variant of KF for tracking multiple narrowband sound sources in reverberant rooms. Operating directly on microphone measurements, it enables continuous 3-D tracking and avoids the limitations of existing methods that rely on discrete grid-based SSL estimates at each time frame. Simulation results show that PNEKF accurately tracks multiple moving narrowband sources under reverberant conditions across diverse frequencies and trajectories. Future work will focus on: (1) extending the framework to broadband sources using a broadband Point Neuron Learning model to reduce microphone requirements; (2) generalization to an unknown number of sources; (3) eliminating SSL-based initialization for fully measurement-driven tracking; and (4) real-world experimental validation.

## REFERENCES

- [1] A. Privitera, F. Fontana, and M. Geronazzo, "On the effect of user tracking on perceived source positions in mobile audio augmented reality," in *Proc. Biannu. Conf. Ital. SIGCHI Chapter*. ACM, 2023, pp. 1–9.
- [2] Y. Zhang, C. Zhao, and L. Guo, "Intelligent video surveillance system based on sound source localization technology," *App. Mech. and Mat.*, vol. 511–512, pp. 530–535, 2014.
- [3] J. Valin, F. Michaud, and J. Rouat, "Robust localization and tracking of simultaneous moving sound sources using beamforming and particle filtering," *Robot. and Auton. Syst.*, vol. 55, no. 3, p. 216–228, 2007. [Online]. Available: <http://dx.doi.org/10.1016/j.robot.2006.08.004>
- [4] D. Desai and N. Mehendale, "A review on sound source localization systems," *Archives of comp. methods in eng.*, vol. 29, no. 7, pp. 4631–4642, 2022.
- [5] K. Nakadai, K. Itoyama, K. Hoshiba, and H. Okuno, "Music-based sound source localization and tracking for tasks 1 and 3," in *Proc. LOCATA Challenge Workshop-Satell. Event IWAENC*, 2018, pp. 1–5.
- [6] Y. Jing, Z. Li, and C. Liu, "Acoustic source tracking based on adaptive distributed particle filter in distributed microphone networks," *Sig. Process.*, vol. 154, pp. 375–386, 2019.
- [7] X. Qian, A. Cavallaro, A. Brutti, and M. Omologo, "Locata challenge: Speaker localization with a planar array," *CoRR*, vol. abs/1901.08983, 2019.
- [8] Z. Liang, X. Ma, and X. Dai, "Robust tracking of moving sound source using multiple model kalman filter," *Appl. Acoust.*, vol. 69, no. 12, pp. 1350–1355, 2008.
- [9] R. Lebarbenchon, E. Camberlein, D. Carlo, C. Gaultier, A. Deleforge, and N. Bertin, "Evaluation of an open-source implementation of the srp-phat algorithm within the 2018 locata challenge," *arXiv preprint arXiv:1812.05901*, 2018.
- [10] D. Salvati, C. Drioli, and G. L. Foresti, "Localization and tracking of an acoustic source using a diagonal unloading beamforming and a kalman filter," *arXiv preprint arXiv:1812.01521*, 2018.
- [11] S. Kitić and A. Guérin, "Tramp: Tracking by a real-time ambisonic-based particle filter," in *IEEE-AASP Challenge on Acoust. Source Localizat. Track.*, 2018.
- [12] Y. Liu, W. Wang, and V. Kilic, "Source localization in reverberant rooms using sparse modeling and narrowband measurements," *arXiv preprint arXiv:1812.01570*, 2018.
- [13] S. Xu, W.-T. Lai, Y. Zhang, J. Zhang, A. Bastine, P. Samarasinghe, and T. Abhayapala, "Three-dimensional gradient-based tracking of multiple sound sources," in *Asia-Pacific Sig. Inf. Proc. Assoc. Annu. Summit Conf. (APSIPA ASC)*, 2025, pp. 113–118.
- [14] H. Bi and T. Abhayapala, "Point neuron learning: a new physics-informed neural network architecture," *EURASIP J. Audio, Speech, Music Process.*, vol. 2024, no. 56, Nov. 2024.
- [15] S. Koyama and L. Daudet, "Sparse representation of a spatial sound field in a reverberant environment," *IEEE J. Selected Topics Signal Process.*, vol. 13, no. 1, pp. 172–184, 2019.
- [16] R. Kalman, "A new approach to linear filtering and prediction problems," *Trans. ASME J. of Basic Eng.*, vol. 82, no. Series D, pp. 35–45, 1960.
- [17] R. Brown, *Introduction to random signal analysis and Kalman filtering*. J. Wiley & Sons, 1983.
- [18] Q. Li, R. Li, K. Ji, and W. Dai, "Kalman filter and its application," in *Int. Conf. on Intell. Net. Intell. Syst. (ICINIS)*, no. 8, 2015, pp. 74–77.
- [19] J. Allen and D. Berkley, "Image method for efficiently simulating small-room acoustics," *J. Acoust. Soc. Amer.*, vol. 65, pp. 943–950, 04 1979.
- [20] S. Xu, J. Zhang, T. Abhayapala, A. Bastine, W.-T. Lai, and P. Samarasinghe, "Sparse sound field representation using complex orthogonal matching pursuit," in *IEEE Int. Conf. on Acous. Speech Sig. Proc. (ICASSP)*, 2024, pp. 1336–1340.
- [21] W.-T. Lai, L. Birnie, T. Abhayapala, A. Bastine, S. Xu, and P. Samarasinghe, "A Two-Step Approach for Narrowband Source Localization in Reverberant Rooms," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. Workshops (ICASSPW)*, Seoul, Korea, Apr. 2024, pp. 490–494.
- [22] C. Knapp and G. Carter, "The generalized correlation method for estimation of time delay," *IEEE Trans. on Acoust., Speech, Sig. Proc.*, vol. 24, no. 4, pp. 320–327, 1976.
- [23] R. Kennedy, P. Sadeghi, T. Abhayapala, and H. Jones, "Intrinsic limits of dimensionality and richness in random multipath fields," *IEEE Trans. on Sig. Proc.*, vol. 55, no. 6, pp. 2542–2556, 2007.